

Proof that $w_{g,t}^S \geq 0$ under Assumption 6 and if $N_{g,t}/N_{g,t-1}$ does not depend on g

Under Assumption 6, one has that $D_{g,t} = 1\{t \geq a_g\}$, with $a_g \in \{1, \dots, T+1\}$. Therefore, given the form of $w_{g,t}^S$, we just have to prove that for all g ,

$$\sum_{t \geq a_g} N_{g,t} \varepsilon_{g,t} \geq 0. \quad (28)$$

Because $N_{g,t}/N_{g,t-1}$ does not vary across g for all $t \geq 2$, we have $N_{g,t} = N_{g,0} \gamma_t$ for some $\gamma_t \geq 0$. Moreover, $\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot}$. Let $\tilde{\gamma}_t = \gamma_t / \sum_{t \geq 0} \gamma_t$, then $D_{g,\cdot} = \sum_{t \geq a_g} \tilde{\gamma}_t$, and $D_{\cdot,\cdot} = \sum_{t \geq 0} \tilde{\gamma}_t D_{\cdot,t}$. Hence,

$$\begin{aligned} \frac{1}{N_{g,0} \sum_t \gamma_t} \sum_{t \geq a_g} N_{g,t} \varepsilon_{g,t} &= D_{g,\cdot} (1 - D_{g,\cdot} + D_{\cdot,\cdot}) - \sum_{t \geq a_g} \tilde{\gamma}_t D_{\cdot,t} \\ &= D_{g,\cdot} \left(1 - D_{g,\cdot} + \sum_{t < a_g} \tilde{\gamma}_t D_{\cdot,t} \right) - \left(\sum_{t \geq a_g} \tilde{\gamma}_t D_{\cdot,t} \right) (1 - D_{g,\cdot}). \end{aligned} \quad (29)$$

Now, because $D_{\cdot,t} \leq 1$,

$$\sum_{t \geq a_g} \tilde{\gamma}_t D_{\cdot,t} \leq \sum_{t \geq a_g} \tilde{\gamma}_t = D_{g,\cdot}.$$

Hence, in view of (29),

$$\frac{1}{N_{g,0} \sum_t \gamma_t} \sum_{t \geq a_g} N_{g,t} \varepsilon_{g,t} \geq D_{g,\cdot} \sum_{t < a_g} \tilde{\gamma}_t D_{\cdot,t} \geq 0.$$

Therefore, (28) and the result follows.

Proof that $w_{fd,g,t}^S \geq 0$

We just have to focus on the cases where $D_{g,t} \neq D_{g,t-1}$. Note that $\varepsilon_{fd,g,t} = D_{g,t} - D_{g,t-1} - (D_{\cdot,t} - D_{\cdot,t-1})$. Then, if $D_{g,t} - D_{g,t-1} = 1$, the numerator of $w_{fd,g,t}^S$ has the same sign as $1 - (D_{\cdot,t} - D_{\cdot,t-1})$, which is positive. If $D_{g,t} - D_{g,t-1} = -1$, the numerator of $w_{fd,g,t}^S$ has the same sign as $1 + (D_{\cdot,t} - D_{\cdot,t-1})$, which is also positive. Because the denominator sums terms that are always positive, it is positive as well. The result follows.

Proof of Theorem S3

The reasoning is exactly the same as in Theorem 1, except that we rely on Lemma S3 below, instead of Lemma 1. We thus only prove Lemma S3.

Lemma S3 *If Assumptions 1-5 hold and $D_{i,g,t} \in \{0, \dots, \bar{d}\}$.*

$$\begin{aligned} &E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t'} | \mathbf{D}) - (E(Y_{g',t} | \mathbf{D}) - E(Y_{g',t'} | \mathbf{D})) \\ &= D_{g,t} E(\Delta_{g,t}^{ACR} | \mathbf{D}) - D_{g,t'} E(\Delta_{g,t'}^{ACR} | \mathbf{D}) - (D_{g',t} E(\Delta_{g',t}^{ACR} | \mathbf{D}) - D_{g',t'} E(\Delta_{g',t'}^{ACR} | \mathbf{D})). \end{aligned}$$

Proof of Lemma S3

Under Assumption 2, we have $E(Y_{g,t}|\mathbf{D}) = E(Y_{g,t}(0)|\mathbf{D}) + E(Y_{g,t}(D_{g,t}) - Y_{g,t}(0)|\mathbf{D})$. The result follows by decomposing similarly the three other terms $E(Y_{g,t'}|\mathbf{D})$, $E(Y_{g',t}|\mathbf{D})$, and $E(Y_{g',t'}|\mathbf{D})$, using Assumptions 3-5, and finally using the definition of $\Delta_{g,t}^{ACR}$.

Proof of Theorem S4

The proof relies on the following lemma, that resembles Lemma 1 and that we do not prove.

Lemma S4 *If Assumptions 1, 2, and S3-S4 hold, for all $(g, g', t, t') \in \{1, \dots, G\}^2 \times \{1, \dots, T\}^2$,*

$$\begin{aligned} & E(Y_{g,t} - X'_{g,t}\gamma | \mathbf{X}, \mathbf{D}) - E(Y_{g,t'} - X'_{g,t'}\gamma | \mathbf{X}, \mathbf{D}) \\ & - (E(Y_{g',t} - X'_{g',t}\gamma | \mathbf{X}, \mathbf{D}) - E(Y_{g',t'} - X'_{g',t'}\gamma | \mathbf{X}, \mathbf{D})) \\ & = D_{g,t}E(\Delta_{g,t} | \mathbf{X}, \mathbf{D}) - D_{g,t'}E(\Delta_{g,t'} | \mathbf{X}, \mathbf{D}) - (D_{g',t}E(\Delta_{g',t} | \mathbf{X}, \mathbf{D}) - D_{g',t'}E(\Delta_{g',t'} | \mathbf{X}, \mathbf{D})). \end{aligned}$$

It follows from the Frisch-Waugh theorem and the definition of $\varepsilon_{g,t}^X$ that

$$E(\hat{\beta}_{fe} | \mathbf{X}, \mathbf{D}) = \frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t}^X E(Y_{g,t} | \mathbf{X}, \mathbf{D})}{\sum_{g,t} N_{g,t} \varepsilon_{g,t}^X D_{g,t}}. \quad (30)$$

Now, by definition of $\varepsilon_{g,t}^X$ again,

$$\sum_{t=1}^T N_{g,t} \varepsilon_{g,t}^X = 0 \text{ for all } g \in \{1, \dots, G\}, \quad (31)$$

$$\sum_{g=1}^G N_{g,t} \varepsilon_{g,t}^X = 0 \text{ for all } t \in \{1, \dots, T\}, \quad (32)$$

$$\sum_{g,t} N_{g,t} \varepsilon_{g,t}^X X_{g,t} = 0. \quad (33)$$

Then,

$$\begin{aligned} & \sum_{g,t} N_{g,t} \varepsilon_{g,t}^X E(Y_{g,t} | \mathbf{X}, \mathbf{D}) \\ & = \sum_{g,t} N_{g,t} \varepsilon_{g,t}^X E(Y_{g,t} - X'_{g,t}\lambda | \mathbf{X}, \mathbf{D}) \\ & = \sum_{g,t} N_{g,t} \varepsilon_{g,t}^X (E(Y_{g,t} - X'_{g,t}\lambda | \mathbf{X}, \mathbf{D}) - E(Y_{g,1} - X'_{g,1}\lambda | \mathbf{X}, \mathbf{D}) \\ & \quad - E(Y_{1,t} - X'_{1,t}\lambda | \mathbf{X}, \mathbf{D}) + E(Y_{1,1} - X'_{1,1}\lambda | \mathbf{X}, \mathbf{D})) \end{aligned}$$

The first equality follows from (33). The second follows from Equations (31) and (32). Hence,

$$\begin{aligned}
\sum_{g,t} N_{g,t} \varepsilon_{g,t}^X E(Y_{g,t} | \mathbf{X}, \mathbf{D}) &= \sum_{g,t} N_{g,t} \varepsilon_{g,t}^X (D_{g,t} E(\Delta_{g,t} | \mathbf{X}, \mathbf{D}) - D_{g,1} E(\Delta_{g,1} | \mathbf{X}, \mathbf{D}) \\
&\quad - D_{1,t} E(\Delta_{1,t} | \mathbf{X}, \mathbf{D}) + D_{1,1} E(\Delta_{1,1} | \mathbf{X}, \mathbf{D})) \\
&= \sum_{g,t} N_{g,t} \varepsilon_{g,t}^X D_{g,t} E(\Delta_{g,t} | \mathbf{X}, \mathbf{D}) \\
&= \sum_{(g,t): D_{g,t}=1} N_{g,t} \varepsilon_{g,t}^X E(\Delta_{g,t} | \mathbf{X}, \mathbf{D}). \tag{34}
\end{aligned}$$

The first equality follows from Lemma 4. The second follows from Equations (31) and (32). The third follows from Assumption 2. Finally, Assumption 2 implies that

$$\sum_{g,t} N_{g,t} \varepsilon_{g,t}^X D_{g,t} = \sum_{(g,t): D_{g,t}=1} N_{g,t} \varepsilon_{g,t}^X. \tag{35}$$

Combining (30), (34), (35), and the law of iterated expectations yields the result.

Proof of Theorem S5

Reasoning as in the proof of Theorem 3, we get that for all $t \geq 2$,

$$\begin{aligned}
N_{d,d',t} E(\text{DID}_{d,d',t} | \mathbf{D}) &= \sum_{g: D_{g,t}=d', D_{g,t-1}=d} N_{g,t} E(Y_{g,t}(\max(d, d')) - Y_{g,t}(\min(d, d')) | \mathbf{D}) \\
&= \sum_{(i,g): D_{g,t}=d', D_{g,t-1}=d} E(Y_{i,g,t}(\max(D_{g,t}, D_{g,t-1})) - Y_{i,g,t}(\min(D_{g,t}, D_{g,t-1})) | \mathbf{D}).
\end{aligned}$$

For all (g, t) , there exists one $(d, d') \in \mathcal{D}^2$ such that $D_{g,t} = d'$ and $D_{g,t-1} = d$. Hence,

$$\begin{aligned}
&\sum_{t=2}^T \sum_{(d,d') \in \mathcal{D}^2: d \neq d'} N_{d,d',t} E(\text{DID}_{d,d',t} | \mathbf{D}) \\
&= \sum_{t=2}^T \sum_{(d,d') \in \mathcal{D}^2} N_{d,d',t} E(\text{DID}_{d,d',t} | \mathbf{D}) \\
&= \sum_{(i,g,t): t \geq 2} E(Y_{i,g,t}(\max(D_{g,t}, D_{g,t-1})) - Y_{i,g,t}(\min(D_{g,t}, D_{g,t-1})) | \mathbf{D}).
\end{aligned}$$

The result follows by definition of DID_M and δ^{SCR} , and the law of iterated expectations.

Proof of Theorem S6

1. Asymptotic normality

Let us define $P^S = N_S/G$ and

$$T^S = \frac{1}{G} \sum_{(g,t):t \geq 2} N_{g,t} \mathbb{1}\{D_{g,t} \neq D_{g,t-1}\} [Y_{g,t}(1) - Y_{g,t}(0)].$$

We prove the result in two steps. First, we prove that $\sqrt{G}(\text{DID}_M - E(T^S)/E(P^S))$ is asymptotically normal. Second, we show that the difference between $E(T^S)/E(P^S)$ and δ^S is asymptotically negligible.

Convergence of $\sqrt{G}(\text{DID}_M - E(T^S)/E(P^S))$

By Assumption S9,

$$\sup_{g,t} N_{g,t}^4 E[\mathbb{1}\{D_{g,t} = d, D_{g,t-1} = d'\} (Y_{g,t} - Y_{g,t-1})^4] < +\infty.$$

Thus, Lyapunov's condition for the central limit theorem holds, and because $\Sigma = \lim_g G \times V(\mathbf{U})$ exists,

$$\sqrt{G}(\mathbf{U} - E[\mathbf{U}]) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

By Assumption S9, $P_{d,d',t}^\infty = \lim_{G \rightarrow \infty} E(P_{d,d',t})$ exists. Then for $d \in \{0, 1\}$, define $\mathcal{T}_d = \{t : P_{1-d,d,t}^\infty > 0\}$ and the event $\mathcal{D} = \{N_S > 0\} \cap \mathcal{D}_0 \cap \mathcal{D}_1$, with

$$\mathcal{D}_d = \{t \in \mathcal{T}_d \iff \min(N_{d,d,t}, N_{1-d,d,t}) > 0\}.$$

By the law of large numbers and Assumption S10, $N_S > 0$ with probability approaching one. Next, fix $d \in \{0, 1\}$. If $t \in \mathcal{T}_d$, then, by the law of large numbers, $N_{1-d,d,t} > 0$ with probability approaching one. Moreover, for such a t , there exists g such that $\Pr(D_{g,t} = 1-d, D_{g,t-1} = d) > 0$ and thus, by Assumption S10, $P_{d,d,t}^\infty > 0$. Then, by the law of large numbers again, $N_{d,d,t} > 0$ with probability approaching one. Conversely, if $\min(N_{d,d,t}, N_{1-d,d,t}) > 0$, then there exists g such that $\Pr(D_{g,t} = 1-d, D_{g,t-1} = d) > 0$. Hence, by Assumption S10 again, $P_{1-d,d,t}^\infty > 0$. This shows that $\mathcal{D}_d$, and thus $\mathcal{D}$, holds with probability approaching one.

Now, by definition of DID_M , under $\mathcal{D}$ we have $\text{DID}_M = f(\mathbf{U})$, with

$$f(p_{0,0,1}, q_{0,0,1}, \dots, p_{1,1,T}, q_{1,1,T}) = \frac{\sum_{d=0}^1 (-1)^d \sum_{t \in \mathcal{T}_d} q_{1-d,d,t} - (p_{1-d,d,t}/p_{d,d,t}) q_{d,d,t}}{\sum_{t=1}^T p_{1,0,t} + p_{0,1,t}}, \quad (36)$$

for all $(p_{d,d',t})_{(d,d',t)}$ such that all denominators are strictly positive. By Assumption S9 again, $E[\mathbf{U}]$ converges to $\mathbf{U}^\infty$. Furthermore, f is continuously differentiable in a neighborhood of $\mathbf{U}^\infty$. Thus, by the uniform delta method (see, e.g. van der Vaart, 2000, Theorem 3.8),

$$\sqrt{G}(\text{DID}_M - f(E[\mathbf{U}])) = J_f(\mathbf{U}^\infty) \times \sqrt{G}(\mathbf{U} - E[\mathbf{U}]) + o_P(1). \quad (37)$$